Active Inference BookStream 002.01 ~ Parr, Pezzulo, Friston ~ Chapters 1,

BookStream | Parr, Pezzulo, Friston ~ Chapters 1, 2, 3, 6 | 1:26:45

All right. Hello and welcome everyone. This is active book stream number 2.0. We are going to be discussing the textbook, active inference, the free energy principle in mind, brain and behavior 2022 textbook by Thomas Parr, Giovanni Pizzullo and Carl Friston. In this live stream with Ali and I, we're going to be giving some overviews and some kind of points that you'll want to have in mind, a little bit of background and context, specifically on chapters 1, 2, 3 and 6. We're going to do more live streams in the future that will cover the other chapters. There's 10 chapters overall. And the goal of this live stream is to use the live stream format to produce materials that we can then clip out so that the textbook groups, which we have multiple ongoing cohorts at the Institute, so that those textbook groups can be really engaging and discussion oriented. Because after several cohorts of going through this, Ali and I and others have started to kind of hone our understanding of the textbook. And so we are starting to feel ready to put down some background and context .zero videos. And then ideally, people who are participating in the textbook group cohorts will be able to view and think about those videos leading up to a discussion so that when we do have the discussions in the textbook group, that those can be maximally interactive and engaging. So before we jump into chapter one, Ali, do you want to just add anything about the textbook or anything else? Hello, I'm Ali. I'm very happy and excited to be here to discuss the book that we've fallen in love for the last year. And despite the fact that we've been studying and discussing this book for more than a year, there's still a lot to learn and unpack. So yeah, great. All right. So then we will begin in just a few seconds, the chapter one and overview.

All right. This section is going to be the chapter one and overview of the active inference textbook. So shown here are the table of contents. There are 10 chapters and three appendices in this book, and they're separated into part one with chapters one through five and part two, chapters six through 10. Ali, do you want to give any overview thoughts on the organization or structure of the book?

Okay. So can you hear me now? Sorry. Yes. Okay. So part one is basically more aligned in terms of the theoretical construction of the active inference, and it provides some conceptual and theoretical tools to be utilized in the second part, because for the second part, we mostly deal with the practical side of active inference, how to actually implement active inference methods to model some of the systems of interest. So that's basically the main justification behind dividing up these chapters into two. Great. All right. We're going to pass over the short preface written by Carl Friston, but it's worth a read. It's only several pages long. So let's go to chapter one overview. So do you have any opening thoughts on chapter one or where does one even begin with such a book?

Sure. Yeah. For chapter one, I think probably the most central theme of chapter one is about the two different approaches to active inference, namely the high road and the low road. And I think it provides

a really nice perspective to view a kind of a whole picture and how the different components of active inference fit together. So, but also I highly recommend chapter 13 from this book, Andy Clark, and it's, and Andy Clark and his critics, written by chapter 13, I think is entitled Beyond the Desert Landscape, written by Carl Friston, in which he provides more elaborate and a bit philosophical reasoning behind what exactly are low road and high road and what are the aims of each approach and the distinction between those two. So for chapter one, we begin obviously with an introductory paragraph, which sets the stage for the, I mean, all the components that would be discussed in this chapter.
So, but a very important section, I believe, is a section two here, which is how do living organisms persist and act adaptively, which is basically the main question active inference is trying to address.
So if we want to describe active inference in, I mean, minimalistically with only one sentence or two, probably we can say something to the effect that it's a theory that address this very question here.
So why they've opened up this chapter with important question. And then action three, which is again, a continuation of section one, but a bit more specifically about active inference and how active inference is addressing the question provided in the section two. So basically, again, active inference is a modeling framework for modeling the behaviors from first principles. And what is meant by first principles will be elaborated upon in the following chapters, the following sections, and also obviously in the following chapters. But very briefly, it's a kind of modeling framework that tries to somehow use variational principles, such as a free energy principle to construct modeling and mathematical tools that will enable us to describe the dynamical systems of living and even non-living agents in terms of those variational principles and how those agents or those systems are coupled with their environment. So that's basically the unificatory principle that is at the heart of this active inference approach, because yes, exactly here, it's a kind of description of this action perception loop that is at the heart of active inference or FEP framework.

And then we go to chapter, sorry, section 1.4, which is again, a description of the structure of the book and why those two parts are divided as such.

So, as I said earlier, first part is more concerned with the theoretical underpinnings of active inference,

and the second part is more concerned with the theoretical side of it, as we see in the subsection headings here.

So that's, yeah, and then a very important diagram, which will provide very helpful as we whole book is provided in figure 1.2,

which kind of ties all the different strands of various related theories, such as predictive coding, predictive processing,

Bayesian brain hypothesis, and so on, into a kind of unified and integrated framework by using those two trajectories of constructing active inference,

namely the high road and the high road.

So it literally here is represented as the high road and the low road in this diagram.

So very briefly,

high road is a kind of top down approach to active inference.

It begins with the question,

how,

how things should act if they exist?

In other words,

in other words,

in other words,

it kind of sets,

sets the stage for developing the concept of Markov blankets.

And then obviously with by setting that also surprise minimization,

it attempts to elaborate that road by adding other aspects of modeling,

other aspects of theory,

such as predictive processing,

self-evidencing,

autopoiesis,

and so on,

and then reach an integrated version of active inference,

which is basically an integration between perception and action.

So I believe the main thing that sets active inference apart from all the other related theories,

such as predictive processing and predictive coding,

is the very deep integration of action and perception.

So action is not just some afterthought or some additional component that needs to be accounted for.

It is deeply woven into the fabric of the theory.

So as we'll see,

especially in chapters two and three,

even the mathematical formalism for perception and action are interestingly similar to each other and even symmetric.

And it shows that perception and action are actually not separate and distinct,

totally distinct entities and concepts.

And active inference definitely provides a very promising framework to integrate them into a holistic framework.

And another approach to construct that same theory is to begin with the probabilistic description of Bayes' theorem

and try to build up the formalism of active inference using Bayes' inference theorem and Bayesian statistical inference.

So again, we here see some very, I mean, apparently different and distinct strands,

such as predictive coding, Bayesian brain hypothesis, and so on.

But as we see, all of those different strands can be united and tied together within this low-road approach to active inference.

So, yes, that's...

I'll just give one short thought.

Thanks. These are great.

Yeah, just one short thought.

Thank you, Ali, for these great summaries.

The low-road and the high-road are going to be explored in a lot more detail in the coming chapters two and three.

We can think of the high-road as being a why because it describes how persistent or remeasurable systems exist.

Now, they might be using Bayes' theorem or they might be using any other kind of mechanism internally,

but the high-road is describing the why of existence.

From the low-road, the bottom-up approach is kind of like a how.

So, you could use Bayes' theorem to describe persistent autopoetic entities,

or you could use Bayes' theorem for other purposes, too.

So, where these two roads intersect is active inference,

and we have a lot more discussion coming and, of course, unpacked in the textbook groups

about how these really map on to bottom-up and top-down causation

and the plurality of whys that we approach in different settings.

Good to continue?

Yes.

Sure.

Some of the coming sections of this chapter, continuing from page eight,

are going to describe what are happening in the next chapters.

So, chapter two is going to set out the low-road perspective.

Chapter three is going to describe the high road.

Chapter four, we will unpack active inference more formally.

That's one of the math-heavy chapters.

Chapter five, we'll move from formal treatments to biological implications of active inference

with a special focus on mammalian neurophysiology.

Chapter five sets out the process theory associated with active inference

and gives a lot of hands-on and empirical examples.

And that's a lot of the first part of the book.

They're then going to summarize some of the key points and distillations from that theory-heavy first section.

One of them is that, as Ali mentioned, perception and action are complementary processes

and ways that fulfill the same imperative, free energy minimization.

We sometimes summarize that by saying that in the pursuit of the minimization of free energy, agents can either change their mind or change the world.

Change your mind is associated with perception and learning,

and those are distinguished later.

And changing the world is associated with action.

There's some discussion about action selection and optimal policy selection.

And all cognitive operations in active inference are conceptualized as inference over generative models.

So the true kernel, the cognitive kernel of active inference, is going to be the generative model.

Want to pick up from here or give any thoughts on generative models and then carry on.

Thanks, Ali.

Just one point is there is a common misconception about generative models,

which is that generative models only refers to the inner states of the system and also its markup blanket.

But actually, generative model kind of encompass all the states of interest of the situation we're trying to model.

So it also encodes the probabilistic information about the external states as well.

So generative model can be thought of as kind of the whole state of the system.

And by system here, I mean the coupled dynamics between the internal and external states.

So that might be helpful to keep in mind.

Great.

Great.

One other distinction that comes up and is used a lot nowadays, although it's not always used in prior literature,

is the distinction between the generative model and the generative process.

We're going to come back to it.

But usually the generative model is being used to describe our statistical model of the agent of interest.

And the generative process is the process that generates observations that are then passed to the generative model as observations.

However, because generative models can be generative processes for each other,

these two are not necessarily distinct in principle.

So in different simulations and in different specific cases,

generative models can also be generative processes for each other.

That leads us towards talking about ecosystems of shared intelligence and interactions amongst generative models.

But when we're talking about generative model, it's like the organism or the system of interest that we're focused on.

There are some more themes that arise from the first half of the book.

Now we're on page 11.

Action is quintessentially goal-directed and purposive.

We talk a lot about how preference plays a role in shaping action selection and active inference.

And the goal-directed nature of active inference will be unpacked in chapters 2 and 3.

They discuss how various constructs of active inference have plausible biological analogs in the brain.

Once one has defined a specific generative model for a problem at hand,

one can move from active inference as a normative theory to active inference as a process theory,

which makes specific empirical predictions.

And there's many, many interesting philosophy of science type discussions that we have about

explain, predict, design, control.

And then we get to section 1.4.2, giving overview on part 2 active inference in practice.

So I'll just continue on.

Chapter 6 is going to introduce a recipe for building active inference models.

So this is where the step-by-step comes into play.

It's going to give a sequence of steps and approaches that you can take to go from understanding the structural aspects of a system or phenomena of interest

and build your way towards having an active inference model that can then be implemented in code.

And we have various examples.

And we're there in the textbook group to work with everyone who wants to take this journey.

But chapter 6 is going to describe that process of building the active inference generative model.

So that's a recipe chapter.

And there's a lot more that goes into the restaurant also as we kind of have fun exploring.

Chapters 7 and 8 are like a pair.

And they describe two different broad families of generative models.

Now, these two families of generative models, they work well, they play well with each other,

but they are very educationally relevant to also consider separately.

Chapter 7 is going to focus on the discrete time generative model.

And chapter 8 is going to focus on the continuous time generative models.

And again, these discrete and continuous time models can be nested and interoperable.

However, they do describe some very important different situations.

And so chapter 7 and 8 have a lot of technical information on the discrete and continuous time formulations of active inference

and also about how they integrate together.

Chapter 9 is illustrations of how active inference models can be used to analyze data from behavioral experiments.

And so this kind of goes beyond chapter 6's recipe.

And it talks about some of the process of making a behavioral experiment

and then using the outcomes of that behavioral experiment, the data from that experiment,

and using that to parameterize and do statistics with the active inference generative model.

And chapter 10 brings it all together.

Chapter 10 is a lot like chapter 1.

And we've even joked of reading the book backwards.

That's how great chapter 10 is.

So for those who are reading chapter 1, consider reading chapter 10 next

and then picking back up with chapter 2

because chapter 10 is going to give some judgments

and also sketch some exciting future directions for where active inference is

in relationship to other fields and where it's going.

So...

I also second that.

Yeah.

The second part of the book illustrates a broad variety of models of biological and cognitive phenomena.

So it's an application-oriented second half of the book.

And we have a lot of accessory code and approaches and notebooks.

So one may not find that this is the one-stop shop that they were looking for.

However, for those who dive in, there's absolutely more than enough to scaffold their learning journey.

And then section 1.5 is the summary.

The chapter introduces the active inference approach to explain biological problems from a normative perspective

and previews some implications of this perspective that will be unpacked in later chapters.

It describes the structural outline of the book,

chapters 1 through 5 being theory, chapters 6 through 10 being practice.

And then it signals that the next two chapters are going to develop the low and the high road perspectives.

And altogether, that is chapter 1.

Any other thoughts on chapter 1?

Not specifically, but I think it provides really good context to the rest of the book

in terms of how all the different elements of active inference comes into play in addressing different problems.

Awesome.

On to chapter 2.

The low road to active inference.

So, all chapters begin with a short quotation.

And the quotation here reads,

My thinking is first and last and always for the sake of my doing.

William James.

So, even before reading the chapter, these quotations are often great places to jump off and have a discussion.

So, Oli, please begin and go as long as you want on the low road to active inference.

Okay.

So, yes.

Chapter 2, as the title suggests,

It's the construction of active inference theory from the viewpoint of the low road to active inference.

So, it begins by providing this notion of perception as inference.

Because more often than not, we usually think of perception as something just in a computational sense of the sense is the processing of the inputs.

Or rather, it's the raw processing of the inputs.

But here it shows that it can also be described as, or even more accurately described, as a kind of inference.

So, it's not just a simple and raw processing of the inputs as my computational model or a computer analogy might suggest.

It's more like something that we predict and then we compare our inputs or the inputs we get from the stimuli with our prediction and then try to somehow minimize that prediction error.

So, I believe that this notion of perception as inference is the most central notion of all the related theories of predictive coding, predictive processing, Bayesian brain hypothesis.

So, up to now, active inference is not very different from all the other theories.

It's basically a subset or a variant of those theories.

But then it progresses into distinguishing the active inference from all the other theories and how it stands apart from the other ones.

So, in section 2.1, we began with some of the basics of probability theory, namely the Bayes theorem in box 2.1.

And some simple examples about how Bayes theorem can be applied.

And again, a very important page or important part of this section will be page 20, which describes the concept of surprisal.

And specifically, the statistical surprisal and how it differs from the phenomenological surprise or psychological surprise.

So, how we can formulate the statistical surprise.

Yes.

Yes.

Just to catch up here.

This example that's going to come back again and again is a person who's guessing the object that's in their hand.

And it could be a frog or an apple.

And the object is going to jump or not.

And so, that's used as a way to talk about the Bayesian updating that forms the kernel of a variety of Bayesian brain-like models, including active inference.

And then there are the presentation of the exact Bayes.

So, in simple cases, you can compute exactly what you want with Bayes theorem.

However, it's a lot more effective on large data sets to use certain approximations and heuristics that we're going to be discussing.

And then, Ali has highlighted that there are two concepts that are very closely related to each other regarding surprise.

There's surprise by itself.

And then there's Bayesian surprise.

So, please pick up there.

What are surprise and Bayesian surprise and why do they matter here?

Yes.

So, well, surprise, I mean, the regular surprise, as we're all familiar with, is something more related to our sense of surprise or psychological sense of surprise of observing some unexpected phenomena or unexpected behavior.

But Bayesian surprise or statistical surprise is, of course, of course, closely related with the psychological sense of surprise.

But a bit more rigorous in which it's a way to compare two probabilistic information using callback Leibler divergence.

And somehow getting the, I mean, how unexpected that information emerging from callback divergence.

And they call it the surprise.

And they call it the surprise.

So surprise is, in other words, is a way to formulating those unexpected probabilistic information.

And it doesn't necessarily align or maps unto perfectly our psychological sense of surprise.

But as I said, it's closely related with that.

Yes.

Surprise.

So, by surprise.

Yeah, please.

Oh, sorry.

I mean, for almost.

Yes.

No, you take it, Ali.

I know.

I was just going to say that.

I was just going to say that by surprise, almost in the rest, all the other sections and chapters of the book, we basically mean this Bayesian surprise sense of the world.

And in order to distinguish that with the regular sense of surprise, sometimes in the literature, surprise is used to refer to Bayesian surprise.

Yes.

So they're both measured in information theoretic units.

This is all happening in information geometric spaces.

The first concept of surprise is applied to a given single observations.

How surprising is that one observation?

And so in that sense, it's a lot like the Z-score of a data point coming in with respect to a statistical distribution.

So it's like you have a height distribution in a classroom and you measure one person and then you can say, what is the Z-score of that measurement?

Was it right at the center of the distribution with a Z-score of zero or were they two standard deviations higher with a Z-score of two?

So it's kind of like that.

And that's why there's a discussion of probability distributions and their support, which is the X values for which they're defined and surprise function with the fancy I.

And that's going to be a function that helps you compute how surprising each observation is given a parameterization of that distribution.

Whereas this Bayesian surprise is more related to learning.

It has to do with how much updating happens between the prior and the posterior, and that's before and after the observation.

So one could imagine a surprising observation, surprise concept one, that either does or doesn't update the prior into a very different posterior.

So as Ali mentioned, they are not exactly the same, but it's going to be important to understand how they're different.

And that's worked out again in the case of the apple jumping.

Box 2.2 continues with a discussion of expectations.

Now, expectation in everyday parlance might be specifically referring to something in the future, like I expect it to rain tomorrow.

And in statistics, when we talk about the expectation, we're talking about the weighted average or the center of gravity of a distribution.

And that can be in both a discrete distribution, at which point you have a weighted sum, or a continuous distribution, in which case it's an integral.

So you can have an expectation for the humidity tomorrow.

And so that might refer to the center of gravity at a t equals plus one.

But just taken alone, expectation means center of gravity of a statistical distribution, not anticipation.

Section 2.3 is going to describe how some of this how, this low road that we're on, is going to connect to biological inference.

There's more discussion of the generative model and the generative process.

And a little bit of a hint that the generative model captures aspects of the generative process,

which is where we see a lot of the classical cybernetic theorems and concepts like requisite diversity, good regulator theorem, and so on.

However, the generative model does not have to be exactly isomorphic with the generative process.

For example, the generative process, the temperature in the room might be a continuous variable.

But then the generative model might be discrete, only modeling integer-based temperatures,

or might be categorical, like too hot, just right, and too cold.

So there's a lot of articulations that can be done because of how flexible and interoperable generative models are with each other.

Figure 2.2 is going to expand upon that chapter one representation of the cybernetic action perception loop.

And we're going to see this more in terms of a generative model and generative process articulation.

And there are incoming, speaking from the perspective of the generative model, the agent,

there are incoming and outgoing dependencies in this graph.

And this is a little bit like a schema, more so than a formal graph, but also it is like a Bayesian graph,

where nodes are variables and edges are causal relationships.

And so we have the internal states of the model, the external states of the generative process,

and then the blanket states that make internal and external states conditionally independent.

And again, speaking from the perspective of the agent, although there is a symmetry,

we can talk about the incoming sensory signals happening from the observations handed from the process,

passed on to the internal states of the generative model,

and then the outgoing actions that are selected that then can influence the hidden state of the world.

For example, going and turning on the heater to increase the temperature in the room.

And so this action perception loop or particular partition is going to get explored in a lot more detail in the coming sections.

Previous section 2.3 was on perception as inference, and now action as inference is going to be discussed.

And that is where they say the discussion to this point is common to all Bayesian brain theories.

We now introduce the simple but fundamental advance offered by active inference,

which is the extension of this inferential perspective to consideration of action as inference.

Perception and action cooperate to realize a single objective.

Section 2.5 is about minimizing the discrepancy between the model and the world.

We already described that there's two ways for this to happen.

Change your mind and change the world.

That's how the discrepancy can be reduced or managed.

We see a variant of the action perception loop where the agent is making perceptions predictive models of the world

that through perception are being juxtaposed with observations handed from the generative process, the world,

and a discrepancy is realized.

Some non-zero discrepancy.

And then here are those two paths to minimize free energy.

Change beliefs by perception and learning or change the world through action.

Section 2.6 is going to discuss how the exact Bayesian approach described earlier is absolutely spot on

if you have infinite computing resources.

However, we're often interested in rapid or large data sets

where we want to be able to get approximate Bayesian computation

or probably approximately correct computation in a vastly accelerated fashion.

And so that is going to be approached using what's called variational Bayesian inference

that's unpacked in chapter 4.

But it suffices to say that variational Bayesian inference implies substituting two intractable quantities,

the posterior probability and log model evidence,
with two quantities that approximate them but can be computed efficiently,
the approximate posterior Q and a variational free energy.
So it transforms an intractable estimation problem into a highly tractable optimization problem.
Pick up from there, Ali.
Yes, sorry.
I just wanted to point out a couple of things, especially about section 2.3,
because I believe it is one of the crucial sections of this chapter,
and in fact, the whole book,
because it provides some of the justifications of using Bayesian inference
as opposed to some other mathematical techniques,
such as maximum likelihood estimation.
But more important than that,
I think it's this concept of optimality,
which comes into play in almost everywhere in the literature and, of course, in this book.
So there are actually two notions of optimality,
which is not discussed in detail here,
namely Bayesian optimality and Jane's optimality.
But in some of the recent papers on Bayesian mechanics,
Dalton's active ad of L-Max,
where Ramstad and others have shown that those two concepts of optimality
are actually congruent with each other.
So that's one of the reasons that the duality between FEP,
free energy principle,
and constrained maximum entropy principle can be...
I mean, it's one of the justifications
for providing that dual formalism between those two.
So, but another point I wanted to mention here is,
because I've seen that using the word hidden state
can be a bit confusing for some people,
because when we observe something as an observation,
obviously it is not, quote-unquote, hidden, right?
So what a hidden state here refers to
is actually the hidden cause of that observation.
So it's not that the observation itself
is hidden from the observation or it's unobserved.
So that might be a bit confusing
if we don't take into consideration

the exact meaning of the hidden state
or latent state here and in the rest of the literature.
So continuing from section 2.6,
here we see one of the two central equations
of active inference,
which is equation 2.5
for variational free energy.
So it's, as I said,
I mean, understanding this equation
and how each line of its formulation represents
in terms of the, I mean, trade-off between energy entropy
or complexity and accuracy or divergence and evidence
is key to understanding almost everything
in the rest of the book
and in many other literature on active inference.
So this is the perceptual part of active inference.
So variation free energy
is a parameter that
is a notion that parameterizes
the surprisal of our perceptual information
we have about the external states.
And then we'll see in the next section
the related and almost symmetrical formulation
to variational free energy,
namely expected free energy,
which is basically the action part
of the active inference.
So we can see how those two
can somehow be seen
as a kind of unified formalism
but described in alternate expressions.
So, but another thing about equation 2.5
is it may be a bit, I don't know,
daunting to see all the relations
between those three lines of equation
and how we can get from one to the other.
So there are some supplementary materials
that we have developed in the past weeks

in which I think can help
in clarifying how the derivations
of these three lines of equation be done.
So I hope they would be clarifying
in, I mean,
and help to understand
how those three lines relate to each other.
So, but the key point here is
to understand that variational free energy
is not something absolute,
but it's just an upper bound
for the minimization.
So as Daniel just mentioned,
it's untractable to have an absolute amount
for the surprisal to be minimized.
So we need to have an upper bound
in order to make that more tractable
because by Jane's inequality,
as we'll saw in chapter four,
I think,
we can see that how the upper bound
of a surprisal
necessarily provides a condition
for minimizing the precise
or the exact free energy.
And that's the key insight
of equation 2.5
or the notion of variational free energy,
which is to provide this upper bound
instead of the exact amount
of surprisal to be minimized.
Yeah, what I'll add there
is if you knew exactly
how well you should be surprised
by a given data point Y,
then you would have had
the optimal model.
However, that is not tractable.

And so by making a quantity
that's always higher
or an upper bound,
the variational free energy F,
which is a function of broadly Q,
our beliefs,
our variational beliefs,
which are built in a way
that makes them very compositional,
very optimizable,
very interpretable,
and data.
And we can reduce the divergence
here, the KL divergence
with a double line
between Q, our beliefs,
and P, the kind of actuality of it.
And if we can reduce this divergence,
in other words,
minimize the free energy,
then we will come closer and closer
to the true surprise function
and do that in a tractable,
incrementally optimizable way.
So equation 2.5
is going to be
the variational free energy,
different ways
that it can be represented
as it takes in data
and beliefs
about the world, Q.
And then
this is going to be
expanded
into the future
to include action
with G,

the expected free energy.
Now,
there's a lot more
that we can say about this.
There's a lot of technicalities
to go into.
But broadly,
notice that
G,
the expected free energy,
is a functional
of policy,
pi,
because it's only
being evaluated
to select
amongst different
action outcomes,
pi.
And
another important difference
is that
it's going to be
describing
sensory outcomes
that haven't yet happened.
A sort of
what would I
perceive
if I did
A
or what if I did
B?
And it's that kind
of comparison
that allows
the expected free energy
functional here

to be used
in action selection
or policy selection
as inference,
planning as inference.
That section
is expanded upon
and in figure 2.6
we see a very nice
representation
of the expected
free energy equation
and then
how when certain
aspects of this
equation
or situation
are zeroed out
we get certain
other familiar
cases.
For example,
where there's
no epistemic value,
there's no
information to learn,
then you get
ruthless
expected utility
theory.
And conversely,
where there is
no pragmatic
value to extract,
so all outcomes
are equally valid
or preferable,
then you get

things like
InfoMax principle
and optimal
Bayesian design.
And then everything
in between
is the space
that we're
interested in.
And so this
figure 2.6
shows that
the expected
free energy
functional
can be seen
as like a
generalization
of a lot
of other
settings
related to
perception
and action
and planning
amidst uncertainty.
And section
2.9
closes the low
road.
They took
us all the
way to
active
inference
from Bayes
theorem
through the

generative
model
on to
active
inference
and
clarifies
these two
notions of
variational
free energy,
that's the
real-time
perceptual
unfolding,
evidence
lower bound
unsurprisal,
tractable,
optimizable,
and so on.
And F.
And G,
the expected
free energy,
which is able
to do
planning as
inference
or policy
selection
as inference.
Expected
free energy
is fundamentally
prospective
and that
enables

counterfactual
cognition.
Section 2.10
summarizes,
active
inference
is a
theory of
how living
artifacts
underwrite
their existence
by minimizing
surprise or
attractable
proxy to
surprise
variational
free energy
via perception
and action.
And they
motivate that
from a
first principles
Bayes
theorem
starting
place.
any
closing
thoughts
on
chapter
two,
Ali?
I would
just

humbly
suggest
for the
people
who want
to go
through
this
chapter
to
try
their best
to really
understand
specifically
what equation
2.5
and 2.6
represents
and how
how to
use those
equations
to
describe
different
situations
with
some
missing
elements
as well
because
I believe
those
sections
and
particularly

those
equations
are
absolutely
essential
for
understanding
everything
active
inference
related
both
in the
rest
of
this
book
and
in
almost
all
things
and
just
a
last
thought
I'll
give
on
chapter
two
is
this
is
exactly
the
work

that
we
do
in
the
active
inference
textbook
group
we
really
welcome
all
backgrounds
every
single
question
and
uncertainty
you
have
is
beautiful
we
have
a
lot
of
resources
that
Ali
and
others
make
to
make
the
math

approachable
and
rigorous
and
natural
language
descriptions
of
the
equations
and
so
on
so
yes
it's
really
important
to
understand
the
equations
because
after
all
that's
like
the
skeleton
that
gives
meaning
to
our
usage
and
fluency
of

the
active
inference
ontology
which
Ali
and I
are
speaking
right
now
we're
not
just
saying
surprise
is
related
to
this
because
we
felt
it
that
way
there
is
an
underpinning
and
it
is
a
really
interesting
life's
work

to
explore
every
time
so
that's
chapter
two
that's
the
low
road
moving
on
to
chapter
three
the
high
road
to
active
inference
and
so
recall
that
the
high
road
is
like
the
why
so
Ali
please
begin

just
go
as
long
as
you
want
on
through
the
high
road
to
active
inference
okay
so
the
high
road
to
active
inference
begin
with
the
question
about
the
conditions
for the
persistence
of
quote
unquote
things
I mean
what

would we
expect
for
a thing
to behave
if it
persists
through
time
and
that's
why
sometimes
FAP
is
referred
to
as
the
theory
of
every
space
thing
and it's
not
absolutely
everything
but
think
just
in this
sense
of
I
mean
persisting
through

time
and
the
notion
that
allows
for
this
thing
to
act
as it
persists
through
time
is
markup
blanket
so
markup
blanket
is
one
of
the
most
essential
ingredients
of
the
high
road
approach
to
active
inference
and
that's

why
this
section
opens
up
with
describing
how
markup
blanket
allows
to
describe
the
situation
of
interest
through
this
conceptual
and
mathematical
tool
but
one thing
that I
always
point out
in almost
all our
textbook
cohorts
is
in this
section
it
doesn't
I mean

exactly
describe
markup
blankets
rigorously
enough
in terms
of
it's
carving
up
the
I mean
state
space
and so
on
so
one
thing
I
think
can be
clarifying
in
following
all the
discussions
around
markup
blanket
is to
keep
in mind
that
it's
just
a

boundary
in the
state
space
so
it's
not
a
necessarily
spatiotemporal

boundary
although
in many
cases
it manifests
itself as
spatiotemporal
boundaries
such as
the
I don't
know
cell
membranes
or
organelle
membranes
and so
on
but
it's
not
necessarily
the
case
so
I

think
it
can be
helpful
to keep
that
in mind
when
we
everywhere
we see
markup
blanket
but
in a
very
simply
markup
blanket
is
what
allows
for
the
agent
or
the
system
to
be
statistically
separated
from
its
environment
as
is
shown

here
in
figure
3.1
so
basically
markup
blanket
mathematically
is
just
the
I
mean
the
sensory
and
active
states
together
it
consists
of
both
of
those
states
and
it
statistically
separates
what
happen
externally
or
internally
from
each

other
so
in
other
words
internal
states
cannot
observe
or infer
anything
about the
external
states
directly
but only
through
the
markup
blanket
and
that's
why
it's
essential
to
describe
the
systems
in this
way
because
obviously
in
any
sparse
coupled
!

systems
there
isn't
a
possibility
to
observe
the
external
states
directly
but
only
through
those
sensory
and
active
states
and
one
other
thing
that
I
also
believe
can
be a
bit
confusing
in
this
picture
is
that
yes
typo

in
the
active
states
and
sensory
states
because
it's
important
to
observe
that
active
state
only
or
the
flow
in
the
active
states
more
precisely
only
depends
on
internal
and
markup
states
or
blanket
states
and
on the
other

hand
in
sensory
states
only
depend
on
the
external
and
blanket
states
so
mu
and
x
in
those
two
equations
should
be
exchanged
yes
yeah
let me
just
unpack
yes
so
this
is
what's
known
as
the
particular
partition

and
that
terminology
was
brought
to the
fore
by
Carl
Fristen's
famous
monograph
in
2019
and
it's
a little
bit
of a
pun
it's
a
particular
partition
because
this
is
just
one
way
to
partition
agent
from
environment
figure
from
ground

and
we
call
what
it
partitions
out
the
blankets
and
internal
states
as
the
particle
so
we
can
think
about
the
most
inert
least
cognitive
particle
is
like
a
speck
of
dust
doing
brownian
diffusion
but
also
there

are
more
sophisticated
kinds
of
cognitive
particles
that
can
include
world
models
counterfactuals
what
would
happen
if
I
did
this
all
of
those
kinds
of
sophisticated
cognitive
operations
that
we
can
explore
in
active
inference
so
note
the

small
typo
that
Ali
mentioned
internal
states
and also
these
edges
reflect
causal
possibilities
amongst
internal
external
and
blanket
states
blanket
constituting
the
active
you
and
sensory
y
states
so
these
are
map
not
territory
this
is not
a
spatiotemporal

articulation
these
are
like
causal
maps
of
the
world
that
may
have
to
do
with
spatial
temporal
boundaries
but
are
not
simply
that
and
sensory
data
or
sensory
states
flow
on
to
internal
states
internal
states
are
involved

in
action
selection
resulting
in
active
states
which
have
some
causal
consequence
in
the
world
degenerative
process
the
external
states
which
results
in
different
sensory
states
coming
in
again
and
so
the
Markov
blanket
is
what
makes
the

internal
and
the
external
states
conditionally
independent
and
that
can
be
thought
of
as
just
a
no
telekinesis
no
telepathy
clause
the
only
way
the
information
comes
across
internal
and
external
states
which
are
symmetrical
with
each
other

is
through
the
boundary
or
the
holographic
screen
or
the
Markov
blanket
you can
continue
on
Ali
okay
so
the
next
section
is
about
surprise
minimization
and
self-evidencing
so again
this term
self-evidencing
is a
common
term in
active
inference
literature
first proposed
by

Jacob
Howie
so
basically
it refers
to how
a system
or agent
or in other
words
a particular
state
can
gather
the evidence
for
its self-persistence
or self-existence
through time
in other words
by
inferring
the state
of the
environment
and
comparing
that
inference
with its
internal
states
or in
other words
with its
generative
model
it's

basically
a way
I mean
an interpretation
of that
that kind
of inference
is
on the
one hand
there is
this
dynamical
interaction
between
the internal
and external
state
but we
can
somehow
interpret
that
physical
dynamics
as
engaging
in
the
active
inference
or
as
model
for
the
persistence
of the

agent
through time
so that's
basically what
refers to as
self-evidencing
and
then we
go through
the
subsection
3.3.1
which
is
which sets
the stage
for
the recent
formulation
of active
inference
as in
mechanics
which is
to
somehow
to
interpret
the

the
the
the
act
of
surprise
minimization
as

minimizing
the
action
through
the
Hamiltonian
principle
of least
action
which is
a
variational
principle
and by
variational
principle
is
what
what is
meant
here
is
just a
computational
or mathematical
tool
that
allows
for
the
computation
or
the
I mean
specific
derivations
to happen
so

it's
not
identical
to
scientific
theories
or scientific
I don't
know
facts
or
observations
it's
just
a tool
a
principle
mathematical
tool
so
again
one of
the main
misconceptions
about
free energy
principle
is that
it is
unfalsifiable
obviously
if we see
it in
this way
it doesn't
make sense
to say
that a

mathematical
tool
or principle
is
unfalsifiable
because
it doesn't
say
anything
about the
empirical
evidence
of
the
phenomena
we're
talking
about
so
that's
why
here
it's
important
to
understand
how
the
surprise
minimization
can be
seen
as
this
kind
of
variational
principle

and
then
equation
3.1
draws
the
parallel
between
the
surprisal
as defined
in section
in chapter
2
sorry
and this
chapter
so
it
ties up
all the
arguments
provided
in the
previous
chapter
with
this
one
and
how
everything
comes
together
in a
single
formulation
of active

inference
stuff
but
they're
just
the
two
distinct
approaches
to
arrive
at
same
destination
and
then
we
go
also
sorry
the
other
important
equation
and
again
another
key
equation
here
is
equation
3.2
which
is
a
kind
of

sorry
yes
equation
I meant
to say
equation
3.2
not equation
3.1
sorry
my bad
so yes
the
other
section
is
about
the
relations
between
inference
cognition
and
stochastic
dynamics
as
I mean
all the
previous
discussions
around
how
the
active
inference
can be
seen
as a

kind
of
self
evidencing
through
the
variational
principles
such as
FEP
comes
together
to
reframe
the
previous
discussions
we saw
in chapter
two
about
perception
as
inference
and
action
as
inference
and
to
see
the
concepts
of
variational
free
energy
and

expected
free
energy
through
the
lens
of
variational
free
energy
variational
principle
of least
action
so
it
unifies
nicely
all the
material
from
chapters
two
and
three
into
something
that
I mean
not
necessarily
distinct
from each
other
but
just
two
sides

of

awesome

yeah

I find

table

three

one

to be

very

exciting

it

draws

together

statistical

physics

bayesian

information

information

theory

and

cognitive

interpretations

so it's

kind of like

learn one

thing

learn many

things

statistical

physics

has been

talking about

minimization

of variational

free energy

for a long

time

and such

methods
are
absolutely
everyday
and
professionalized
in
bayesian
statistics
active
inference
is using
exactly
just
that
to
describe
perception
and action
and so
on
so that's
very
exciting
box
3.2
describes
free
energy
in
statistical
physics
and
active
inference
I
sometimes
joke

that we
have
a few
kinds
of
free
energy
we
have
Tesla
like
electrical
power
should be
available
to
everybody
for no
cost
we're not
talking
about
that
right
now
there's
Gibbs
free
energy
which
is
the
quantity
that
makes
chemical
or
thermochemical

reactions
irreversible
like
the
hydrolysis
of
ATP
and
when
we're
talking
about
variational
free
energy
we're
talking
about
an
information
geometric
space
with
similar
dynamics
similar
kinetics
and
thermodynamics
but
rather than
describing
the reaction
coordinates
of a
chemical
reaction
we're

thinking
about
it
in
terms
of
Bayesian
updating
and
this
is
all
called
the
Bayesian
mechanics
3.41
continues
on
with
variational
free
energy
3.42
goes into
expected
free
energy
so we
see this
a lot
F
variational
free
energy
that's
the
real-time

unfolding
sensory
flow
and
then
G
expected
free
energy
and
the
policy
planning
as
inference
section
3.5
concludes
with
a novel
foundation
active
inference
to
understand
behavior
and
cognition
and
it
describes
a few
features
of
active
inference
including
its

distinguishing
features
from a few
other ways
that people
have looked
at behavior
and
cybernetics
section
3.6
goes into
a bit
more
detail
on
models
policies
and
trajectories
having
to do
with
agency
and
policy
selection
3.7
reconciliation
of
inactive
cybernetic
and
predictive
theories
under
active
inference

again
the
exact
kind
of
thing
that
there's
a whole
literature
on
and
it's
always
amazing
to hear
everybody's
perspective
on
in the
textbook
groups
3.8
active
inference
from
the
emergence
of
life
to
agency
and
3.9
summary
you want
to just
give any

other
thoughts
that you
have
on
chapter
three
again
it's
one
of
the
probably
most
fundamental
chapters
and
I mean
the topics
covered
in this
chapter
is
essential
are
absolutely
essential
to
understand
everything
active
inference
related
but
specifically
the
discussions
in

section
3.6
onward
again
I believe
is really
important
to understand
the
broader
context
of active
inference
and how
it
relates
to
all the
other
theories
but
specifically
section
3.8
provides
a nice
view
of how
we can
use
the same
modeling
the same
theoretical
tools
to model
both
non-living

dynamical
systems
or sparse
couple
dynamical
stochastic
systems
and also
to
the
systems
that
has
agency
or
sentience
so
it's
a much
broader
theoretical
framework
that doesn't
restrict
itself
to only
particular
kinds
of
systems
or
agents
and
we'll
see
much
more
elaboration

on that
in
the
later
time
wonderful
all right
that concludes
our
overview
on chapter
three
now we're
going to
skip
to chapter
six
so again
this live
stream
is just
providing
some
materials
on chapters
one two
three and
six
eventually
we're going
to get
to all
chapters
and version
them
and have
as many
of you

as want
join into
the process
of constructing
these
but this
is also
that our
textbook
groups
can be
really
interactive
and people
can show
up
having
listened
to these
background
and context
videos
all right
so now
we are
in chapter
six
a recipe
for designing
active
inference
models
Abraham
Lincoln
give me
six hours
to chop
down a

tree
and I
will spend
the first
four
sharpening
the axe
all right
Ali
what does
the quote
mean
and please
lead us
into this
discussion
okay
so as the
opening quotation
suggests
it's about
honing the
skill set
and a
skill set
for applying
active
inference
framework
to
model
the
actual
empirical
situations
that
we want
to

we may
use active
inference
for
and
it
frames it
as a
four-step
recipe
to
actually
do
this kind
of model
although

it's
termed
as
recipe
but
I
believe
it's
more
like
a
guideline
and
it's
not
like
I
don't
know
cooking
recipe

that
needs
to be
observed
strictly
so
it's
more
like
suggestions
or
guidelines
to
begin
to
apply
active
inference
framework
in
any
kind
of
situation
we'd
like
so
section
6.2
outlines
those
four
steps
so
the
first
step
is

which
system
are
we
modeling
in
terms
of
how
we
can
define
the
markup
blanket
that
most
effectively
and
efficiently
the
problem
or
question
we
want
to
explore
through
that
modeling
through
the
modeling
of that
situation
so
first

step
is about
defining
those
boundaries
through
markup
blanket
and
the
second
step
is about
what's
the
most
appropriate
form
for the
generative
model
because
obviously
we can
have
many
different
types
of
generative
models
depending
on
again
what
phenomena
or
what

questions
we're
trying
to
address
the
third
stage
is
how to
set up
the
generative
model
so
after
settling
upon
the
type
of
generative
model
we
want
to
use
we
can
talk
about
the
details
of
the
generative
model
and

how
it
can
be
efficiently
modeled
to be
both
tractable
and
also
useful
to
address
the
situation
of
interest
and
finally
how to
set up
the
generative
process
so
as we
saw
in
chapter
two
there's
an
important
but slight
distinction
between
generative

process
and
generative
model
so
they're
not
always
the
same
but
they're
closely
related
to
each
other
generative
process
the
hidden
states
of
the
external
world
and
generative
model
is
how
the
agent
infer
about
those
hidden
states

provided
by
generative
process
so
considering
that
the
environment
is
always
very
complex
to
be
modeled
precisely
and
exactly
with
every
parameter
accounted
for
we
need
to
have
a
restricted
set
of
parameters
to
account
to
account
for

in
the
generative
process
so
that's
what
stage
four
deals
for
deals
about
so
all
of
these
steps
even
if
not
observed
or
followed
sequentially
are
really
I mean
essential
to
setting up
I mean
or the
initial
setup
of
the
situation

we're
trying
to
model
and
I
cannot
imagine
how
any
scenario
can
progress
without
observing
any of
these
steps
at least
I mean
just
by
thinking
about
how
to
set
up
the
situation
or
scenario
awesome
and
in
the
rest
of

the
rest
of
the
chapter
yes
just
wanted
to
highlight
again
these
steps
don't
have
to
be
followed
in
order
they
really
are
like
a
summary
as
Mark
has
just
written
in
the
live
chat
here
it
really

is
like
a
summary
of
the
earlier
parts
of
the
book
in
terms
of
things
that
you
want
to
capture
in
your
consideration
and
also
it's
a
great
way
to
connect
a lot
of
the
technical
ideas
that
are

brought
up
about
generative
model
generative
process
and
so
on
with
some
bigger
questions
like
who
are
we
why
are
we
making
this
model
how
is
the
model
going
to
be
used
so
we
take
that
in
a

lot
of
different
ways
in
the
textbook
groups
but
which
system
are
what
is
the
most
appropriate
form
for
the
generative
model
how
should
we
set
up
the
generative
model
and
how
to
set
up
the
generative
process

are
all
questions
that
must
be
addressed
at
least
when
actually
coming
through
with
carrying
out
a
theoretical
or
empirical
active
inference
modeling
task
and
the
following
sections
are
going
to
go
into
more
detail
on
each
of

those
four
questions
so
Ali
do you
want
to
go
for
this
6.3
and
carry
on
okay
so
section
6.3
is
the
elaboration
of the
first
step
that we
just
saw
which
I mean
what
systems
are we
modeling
and
by this
question
we mean

how
I mean
what
kind
of
system
I mean
how
the system
carves
up
the
state
space
using
the
markup
blanket
or
more
precisely
the
active
states
and
sensory
states
and
I
believe
it
is
both
one
of the
most
fundamental
and

also
one of
the
most
challenging
steps
to be
carried
out
because
markup
blanket
is just
a conceptual
tool
we use
to
frame
the
natural
phenomena
or any
other
scenario
we want
to
model
so
that
scenario
or that
model
obviously
doesn't care
about how
we set up
our markup
blanket

or how
we need
our markup
blanket
but
deciding
on a
proper
set of
active states
and sensory
states
that
would allow
us to
address
the
problems
we need
to explore
or the
questions
we need
to answer
is not
a trivial
task
and it
needs
lots of
lots of
consideration
and even
creativity
on part of
the
researcher
or

anyone
who wants
to
design
an active
inference
based model
for the
problem
so that's
where
the
significance
of
doing
these
kinds
of
modeling
with
the
insights
and
creativity
of
a
researcher
is
I mean
it will
be obvious
because
it's not
just
something
mechanical
or
despite

the fact
that
the word
recipe
can
somehow
make us
believe
that it's
just a
recipe
to follow
but
in actuality
it's not
the case
at all
so
that's
the discussion
of
markup
blanket
and
what
criteria
we need
to keep
in mind
to settle
upon
a proper
markup
blanket
around
our
system
and

the next
section
what is
the most
appropriate
form
for the
generative
model
so
here
again
we can
have
various
way
to set up
our
generative
model
in terms
of
whether
it's
discrete
or
continuous
variables
we want
to
account
for
about
the
timescale
is it
shallow
or

hierarchical
about
the temporal
depth
of the
inference
and
planning
so
these
are
all
problems
we need
to
be
precise
about
questions
we need
to
precise
about
whenever
we want
to
set up
our
generative
model
and
then
we
go
to
section
6.5
how to

set
up
the
generative
model
so
after
precisely
describing
what needs
to be
accounted
or what
parameters
and
the nature
of the
generative
model
needs to
be
accounted
for
then
we
will
obviously
need
to
address
this
question
of
how
can
that
generative
model

be
actually
set up
so
setting up
the variables
for
the generative
model
will
I mean
we will
be concerned
about
how
those
matrices
as
A,
B,
C,
and D
matrices
can be
defined
as
we saw
in
chapter
4
and
then
which
parts
of the
generative
model
are

fixed
and
what
is
learned
so
again
it's
really
important
to
know
what
are
the
dynamics
of
those
parameters
are
in
terms
of
the
learning
process
and
what
needs
to be
updated
consistently
or
learned
as
opposed
to
the

variables
that
either
don't
require
consistent
updating
or
they
vary
quite
slowly
as
compared
to
the
timescale
of
the
generative
model
and
then
we
go
through
the
process
of
setting
up
the
generative
process
and
as I
mentioned
earlier

it is
probably
one of
the
most
complicated
steps
of the
recipe
because
of the
complexity
the
enormous
complexity
of
any
real
time
situation
and
particularly
the
environment
that
the
agent
will
act
and
behave
but
even
if
not
having
I
mean

even
if
there
is
not
any
strict
or
I
don't
know
obvious
way
to
go
through
this
stage
of
setting
up
the
generative
process
this
section
provides
some
helpful
criteria
or
helpful
points
in
order
to
help
us

in
that
stage
as
well
so
finally
after
going
through
all
of
these
stages
we
need
to
somehow
get
I
mean
obviously
we need
results
and we
need
to
provide
our
results
in
a
we
get
our
results
from
our

active
inference
model
and
section
6.7
addresses
that
I
mean
that
very
idea
of
how
to
simulate
visualize
analyze
and fit
data
using
active
inference
because
it's
not
just
enough
to
set
up
the
model
we
need
to
get

the
model
to
work
so
how
can
we
feed
the
data
into
the
model
how
can
we
read
the
data
and
finally
how
can
we
analyze
the
simulated
data
through
the
active
inference
so
this is
in a
way
a kind

of
preliminary
recipe
or preliminary
procedure
for doing
active
inference
or
active
inference
modeling
or at
least
suggested
way
to
do
those
kinds
of
modeling
but
in the
subsequent
chapters
in particular
chapters
7 and
8
we'll go
through
some
helpful
case
studies
for
how

to
actually
use
these
procedures
to
model
some
real
world
situations
and
so
awesome
thanks
well
there's
a lot
to say
on
chapter
6
one
piece
that
comes
up
a lot
is
what
is
in
the
active
inference
kernel
the
cognitive

kernel
and
what
are
situations
that
we
can
model
with
active
inference
there's
a
huge
variety
of
cognitive
phenomena
and
statistical
outcomes
that
we
might
be
interested
to
study
like
counterfactuals
and
planning
like
multi-scale
attention
covert
action

like
the
complex
ways
in
which
memories
might
influence
decision
making
in the
moment
based
upon
associations
made
between
different
sensory
modalities
every
amazing
real world
situation
you can
imagine
and
it's really
helpful to
think about
active
inference
as more
like a
framework
of
interoperable

motifs
that we can
compose
and indeed
even be
creative with
rather than
giving us all
the answers
at the
core
model
because
the
kind
of
essential
minimal
active
inference
model
doesn't
necessarily
include
every
single
attribute
that you
might be
interested
in
just like
any
given
linear
regression
isn't
going to

include
every
single
feature
that
you're
looking
for
this
is
happening
in a
different
type
of
modeling
framework
but
still
there's
a lot
of
customization
and
development
that
goes
into
adapting
or
elaborating
an
active
inference
type
model
for
your

given
system
or
scenario
of
interest
chapter
six
gives
us
some
helpful
guidelines
and
things
to
consider
which
we've
unpacked
and
expanded
on
elsewhere
and
in
the
textbook
group
we
work
together
to
characterize
different
systems
of
interest

to
map
different
parts
or
different
observables
from
that
system
using
the
active
inference
ontology
into
a
unified
active
inference
type
model
so
chapter
six
starts
the
second
half
of
the
book
which
is
the
more
practice
oriented

part
of
the
book
and
the
rest
of
the
part
two
is
going
to
be
a
lot
more
about
the
specific
kinds
of
generative
models
that
you
can
use
discrete
and
continuous
time
and
about
using
them
with

empirical
data
in
chapter
nine
but
chapter
six
just
stands
!
what
is
the
most
appropriate
form
for
the
generative
model
how
to
set
up
the
generative
model
and
how
to
set
up
the
generative
process
so
that

we
can
be
active
inference
modelers
ourselves
all
right
well
Ali
thanks
so
much
for
all
of
these
great
conversations
and
cohorts
we
will
be
able
to
process
this
video
to
clip
out
these
first
versions
of
the

background
and
context
overviews
and
we'll
distribute
them
to
the
cohort
so
that
people
can
show
up
with
their
curiosities
and
passions
and
we'll
see
where
that
goes
any
last
thoughts
thank
you
it
was a
pleasure
to
go

through
these
chapters
once
again
with
each
time
obviously
with
a bit
more
understanding
than
the
previous
one
so
I'm
looking
forward
to
the
next
cohorts
awesome
all right
thanks
Ali
farewell
to

Thank you.

Thank you.